Idea 2: Per-Context Thresholds for AI-Agent Monitors

Rounds 1–4 after three audits: heterogeneity, allocation rules, transfer, poisoning, white-box probes and severity tiers

October 23, 2026 (supersedes the October 22 version)

Summary. Honest monitor-score distributions differ across deployment contexts, so one global threshold over-audits some contexts and under-audits others; Radar-style CFAR holds the false-alarm rate constant in every context. Four rounds tested CFAR and alternatives at a fixed *global* audit budget B on real monitor streams (Gemini-3.1-Pro on 7 settings and ControlArena, LinuxArena, MRT, MonitoringBench, 14 white-box probe streams) against uniform, context-selecting, probing, poisoning and severity-aware attackers. Numbers reflect three audits and the threshold-convention fix (§2.5); changed claims are listed in Section 8. Tags: [C] confirmatory (pre-specified), [E] exploratory.

1. **Heterogeneity is real across deployments, small within one, large for white-box probes** [E]: per-setting FPR 0–9.0% at a global 5% threshold over 7 settings ($\tau = 1.34$, $p < 0.001$); probe τ up to 3.79 vs ≤ 0.46 (n.s.) for LLM monitors on the same trajectories.
2. **CFAR is the worst rule against an attacker who chooses where to attack** [E]: 7 settings, $B = 1\%$, exact spend matching, -0.280 [-0.46, -0.22] selection safety vs global, and worst at every budget. Three to five alarm-feedback probes per context suffice to exploit it.
3. **Damage-equalizing (Stackelberg) thresholds beat global against selection at 5%** [E] ($+0.076$ [$+0.026$, $+0.124$], 7 settings). **At 1% the pre-specified round-4 test was met but is not robust** [C]: $+0.107$ [$+0.024$, $+0.152$] with trajectory-level resampling, driven by SHADE (5 main tasks); with main-task resampling [-0.003, $+0.251$]. The spend matching picks the operating point on held-out honest data: fair across rules, not deployable. Fit on other side tasks Stackelberg loses 0.13 to global at 1% [E].
4. **Per-context rules have a spend floor** $\approx \sum_c w_c/(n_c + 1)$, so they need roughly $1/B$ calibration runs per context (about 300 at 1%); with ≈110–125 per context (ControlArena, public LinuxArena) they are not estimable at 1%, and the white-box "CFAR doubles probe TPR at 1%" is not estimable [E].
5. **Recalibration can be poisoned** [E]: CFAR breaks at poisoned share $\varepsilon \approx B$; a frozen trusted-reference tail holds (worst-case TPR 0.59–0.81 at $\varepsilon = 10\%$) but over-spends 2.0–2.9× under genuine deployment-wide drift.
6. **Watcher-like fixed severity tiers lose to a plain suspicion threshold** [E]: blind severity AUROC 0.53–0.54; tiers cannot run at 1%; at 5% they lose 0.06–0.12 damage-weighted detection (gpt-4.1-mini permissive -0.082 [-0.191, -0.021]).

Spend: $38.93 OpenRouter over four rounds (round 4 $0); audits and threshold re-run $0.

1 Question and scope

How much do honest monitor-score distributions vary by context (environment, setting, task type, agent model, monitor)? At a **fixed global audit budget** B (the fraction of honest traffic audited), do per-context thresholds beat one global threshold, and which per-context rule?

- **Round 1** (Oct 20): heterogeneity; CFAR vs global vs detection-maximizing vs damage-equalizing (Stackelberg) allocation against uniform and context-selecting attackers; small

samples; poisoning of one context's recalibration data; slow ("boiling frog") poisoning.

- **Round 2** (Oct 21): transfer to new attack distributions, new agent models and new monitors (E1); poisoning spread over several contexts (E2); severity-weighted damage (E3); probing attackers and estimation error (E4); white-box and strong-monitor streams (E5).
- **Round 3** (Oct 21): Watcher-like products that combine per-action suspicion with intent-agnostic severity tiers.
- **Round 4** (Oct 22, track `r4ctx`, pre-specified): allocation at $B = 1\%$ on panels with hundreds of honest runs per context, with an exact spend matcher; the round-3 Watcher tables re-summarized without clamping.
- **Audits** (Oct 21–23): the first audit (C1, M5, m2, m3, m11), the second audit (m9 on the round-3 matcher) and the third audit (round-4 resampling unit and operating point) changed several claims; the threshold-convention re-run changed none here. Section 8 lists them.

Prior expectations. Per-context thresholds are requested by several research agendas (*Salient Directions*, Scale MRT, UK AISI Async Control, GDM's UEBA mention). Pinto (2607.06596) shows that a 1% threshold transferred across model families realizes 0–94% audit spend. Detection theory (Ghafouri et al.) predicts that against a threshold-aware attacker the optimal per-context thresholds equalize worst-case *damage*, not FPR, so CFAR may be exploitable.

Sources. All paths in brackets are relative to `results/` unless they start with `data/` or `EXPERIMENT_LOG`. Round 1: [`t2/tables.md`]; round 2: [`r2_t2/tables.md`]; E5: [`r2_t2/e5_tables.md`] and the corrected [`fixes/r2_t2/e5_tables_corrected.md`]; round 3: [`r3_t2/tables.md`], [`r3_t2/watcher_summary.csv`], re-summarized in [`r4_ctx/r3t2_corrected/`]; round 4: [`r4_ctx/tables.md`], [`r4_ctx/summary.csv`], [`r4_ctx/primary.csv`]; audits: [`fixes/corrections.csv`], [`audit/issues.csv`], [`audit2/issues.csv`], [`audit2/r3t2_matched_check.csv`], [`audit3/corrections.csv`], [`audit3/ctx_primary_checks.csv`]; threshold re-run: [`fixes2/headline_threshold_rerun.csv`].

2 Data, rules and evaluation

2.1 Panels

A *panel* is a dataset × monitor × context key. The trajectory statistic is the maximum of per-step (or per-chunk) scores.

2.2 Rules

2.3 Attackers

- *Uniform*: attacks every context in proportion to traffic. Reported as uniform TPR.
- *Informed selection*: knows the published thresholds and picks the context with the highest undetected-success probability, estimated from its own calibration attacks. Reported as **selection safety** = TPR in the chosen context. Variants: oracle (min test TPR), cell-level (context × attack type), damage-weighted.
- *Probing* (round 2): sees only alarm / no-alarm feedback on q probe attacks per context; probes are either free (staging copy) or costly (any probe alarm catches it).
- *Poisoning*: injects a fraction ε of benign-looking but high-scoring runs into the recalibration window of one, several or all contexts.

Panel (round)	Monitor (scale)	Context (#)	Honest / attack	Honest per ctx
la_env (local / cum.) (1)	Gemini-3.1-Pro streaming, aksh-n LinuxArena gold set (0–10)	environment (10)	1,396 / 1,305	87–197
la_taskcat (1)	same	task category (5)	1,332 / 1,296	75–620
la_model (1)	same	agent model (8)	1,155 / 1,298	45–583
akshn_setting, "7 settings" (1–3)	Gemini-3.1-Pro local, same rubric	setting (7)	5,189 / 5,794	328–1,396
ca_setting_model (1–2)	same	ControlArena setting × agent (12)	2,997 / 3,000	247–250
lasweep_{m} (1–2)	12 LinuxArena-sweep monitors (0–100)	environment (10)	200 / 101–142	20
mrt_{m} (1)	claude-3.5-sonnet sequential $N = 5$, 3 prompts (1–5)	CUA task (15)	150 / 675	10
mb_{m} (1–2)	gpt-4.1-mini / nano, trajectory and per-step	BashArena task cluster (7)	535 / 752–2,584	30–207
E5 (2)	17 signals: 7 Qwen3.5-27B white-box streams × {full, none}; gpt-4.1 none; gpt-4.1-mini none / full	task cluster (7)	550 / 541	30–213
mb_mini, mb_mini_full, mb_strong (3)	suspicion: gpt-4.1-mini none / full, gpt-4.1 none; severity: gpt-4.1-mini Watcher tiers (1–10)	task cluster = "org" (7)	535 / 752; 505 / 143	30–207
akshn_adj (3)	Gemini suspicion + outcome-adjudicated severity (label-leaking)	setting (7)	5,189 / 5,794	328–1,396

Table 1: Panels. The Gemini panels are strong monitors (AUROC ≈ 0.99); the MonitoringBench panels are weak (AUROC 0.59–0.76). E5 attacks are 33 MonitoringBench originals plus 508 T3 monitor-aware refined variants of 148 base attacks; 94% of them have LLM-fabricated tool outputs. Sources: [t2/tables.md], [r2_t2/e5_tables.md], [fixes/r2_t2/e5_tables_corrected.md], [r3_t2/tables.md].

- *Severity-aware* (round 3): picks the (org, damage tier) cell with the highest expected undetected damage.

2.4 Evaluation and the spend-matching problem

- **Splits.** Round 1: 60 stratified 50/50 splits (points) and 150 split-then-bootstrap replicates (95% intervals; paired differences vs global are percentiles of the replicate differences). Round 2: 150 paired bootstrap replicates. E5 corrected: 30 plain + 60 bootstrap splits with the attack split grouped by base attack. Round 3: 40 replicates.
- **Nominal mode**: each rule at its nominal level; per-context rules can over-spend out of sample.
- **Spend-matched mode**: the nominal level is swept and the metrics are interpolated at realized test spend $= B$. Rounds 1–3 swept $B \times [0.25, 2]$ with `np.interp`, which *clamps* when the sweep never reaches B; `matched_ok` marks the share of replicates that were bracketed (audit C1). The fixed matcher (`tracks/fixes/e5/`) sweeps $B \times \{0, 0.01..3\}$ and declares a split unmatched otherwise; rows matched in $< 80\%$ of splits are reported as not estimable (*n.e.*). It was applied to E5; the round-3 replicates were re-summarized with matched-only replicates (audit2 m9). The round-1 and round-2 tables were not re-run: in this report rows with `matched_ok` < 0.8 are marked with † and not interpreted.
- **Exact spend matching (round 4).** For each rule the fit level is bisected until the realized *test* spend $\sum_c w_c \mathrm{FPR}_c$ brackets B, and the two bracketing fits are mixed as a randomized

Rule	Definition (all fit to spend B on calibration honest data, randomized ties)
global	one threshold on the pooled honest scores
CFAR	per-context threshold at level B (same FPR everywhere)
Mondrian	per-context conformal quantile
pooled_eb / pooled_cv	per-context FPR shrunk toward global with prior strength n_0 (empirical Bayes or 5-fold CV)
max-detect	fractional knapsack on per-context ROC hulls, maximizing uniform-attacker TPR
Stackelberg	randomized thresholds on ROC hulls minimizing $\max_c D_c(1-\mathrm{TPR}_c)$; $D_c \equiv 1$ or damage-weighted. Round 2 adds shrinkage of each context's attack hull toward the pooled attacks (pseudo-count m_0 or fraction λ; $\lambda = 1$ gives global)
cell minimax (r2)	thresholds minimizing the worst (context × side task) cell's expected undetected damage, optionally shrunk
OS-CFAR / TM-CFAR	per-context robust location (p75 offset / 10%-trimmed mean) plus a common tail level. Tail sources: leave-one-context-out (LOCO, round 1); frozen trusted reference (os_frozen), median / low quantile / trimmed over contexts, capped, guarded (round 2)
xfer_offset (r2)	per-context normal-score offsets learned on a source (other agents or other monitors), one level recalibrated on the target
Watcher tiers (r3)	*permissive*: tier 9–10 always flagged, 7–8 if suspicion $> h$, 1–6 never; *strict*: 9–10 always, 4–8 if suspicion $> h$, 1–3 never; tier_w / tier_opt: one threshold per tier with FPR weights fixed $(0, 1, 3, 10)$ or tuned
sev_tilt (r3)	one global threshold on $\max_i(s_i + \beta v_i)$, $\beta \in \{0, .25, .5, 1, 2\}$ tuned on calibration attacks

policy so that realized spend equals B exactly (max $|\text{spend}-B| = 3.5 \times 10^{-17}$). A replicate whose spend floor exceeds B is unmatched; rows matched in $< 80\%$ of cells are *n.e.*. Because the operating point is chosen on the test honest half, the comparison is at equal realized spend for every rule but is an oracle: a deployed rule must be fit at a nominal level, and nominal per-context fits over-spend (audit 3).

- **Small samples.** Calibration subsampled to $n \in \{10, 30, 100\}$ honest runs per context.

2.5 Threshold convention

The global rule and per-context quantiles used a plug-in randomized quantile, whose out-of-sample FPR for continuous scores is $(\lfloor \alpha n \rfloor + 1)/(n+1)$ (one order statistic too high at integer αn). The repository now uses the conformal randomized rule (a new score s alarms with probability $\mathrm{clip}((\alpha(n+1) - \#\{S_i > s\})/(\#\{S_i = s\} + 1), 0, 1)$, expected FPR exactly α for exchangeable scores).[audit3/corrections.csv] For this report the change hardly matters, because every headline comparison is spend-matched on the test honest half, which fixes realized spend whatever rule produced the thresholds. The re-run left the round-4 primary unchanged (the main-task interval moved from $[-0.004, +0.251]$ to $[-0.003, +0.251]$) and moved the 5% overview contrasts by ≤ 0.004.[fixes2/headline_threshold_rerun.csv] At nominal (unmatched) level it matters: the conformal rule removes plug-in CFAR's nominal over-spend (7 settings at 5%: 5.11% → 5.00%; ControlArena 5.31% → 5.00%), and a conformal version of the Stackelberg hull lowers the nominal round-4 Stackelberg gain from +0.134 to +0.061 [+0.010, +0.119] (part of the nominal gain is plug-in over-spend). For per-context quantiles at small n_c the conformal rule behaves like Mondrian below: it cannot alarm with certainty when $\alpha(n_c + 1) < 1$. Round 1–3 tables were not re-run and keep plug-in thresholds.

2.6 Confirmatory and exploratory results

Only the round-4 primary contrast (Stackelberg − global selection safety on 7 settings at $B = 1\%$, in distribution, spend-matched; supported iff the 95% joint-bootstrap interval is above 0 and both rules are matched in $\geq 80\%$ of cells) was written down before its data were analysed [C]. The design prose was committed first; the code was committed with the results and matches the prose (audit 3). Every other result in this report, in all four rounds, is exploratory [E] and is reported with its interval and without a multiplicity claim.

3 Round 1: heterogeneity, allocation, small samples, single-context poisoning [E]

3.1 Heterogeneity

Panel	ICC of honest max	Per-step variance share	Per-context FPR at global 5%	τ at 5% (p)
7 settings	0.17 (0.31 normal scores)	0.004	0–9.0%	1.34 (< 0.001)
ControlArena setting × agent	0.28	0.014	0–15.8%	1.75 (< 0.001)
LinuxArena environment (local)	0.05	0.007	2.1–7.2%	0.14 (0.38); 0.45 at 10%
LinuxArena task category (local)	0.04	0.005	3.7–12.8%	0.40 (0.04)
LinuxArena agent model (local)	0.00	0.001	2.1–8.0%	0.19 (0.28)
LinuxArena sweep (12 monitors, 20 honest/env)	0.05–0.31	0.009–0.092	0–25%	0.00–1.46
MRT (3 prompts, 10 honest/task)	0.06–0.46	0.08–0.38	0–49%	0.00–1.94
BashArena clusters (gpt-4.1-mini / nano)	≤ 0.05	≤ 0.003	0–12.3%	0.00–0.92

Table 2: Heterogeneity of honest trajectory-max scores. τ = SD of logit per-context FPR net of binomial noise (logit-normal random effects; LR test). [t2/tables.md]

- Context explains little *per-step* variance (0.1–9%; MRT 8–38%). The heterogeneity lives in the trajectory tails.
- **Agent model is not a useful context** (ICC 0.00, τ n.s.).
- LinuxArena-sweep environment effects are shared across the 12 monitors: mean pairwise Spearman 0.70 (min 0.43); common-effect variance 0.148, environment×monitor interaction 0.000, cell noise 0.037. Some environments are intrinsically sensitive-looking for every monitor. [t2/tables.md]

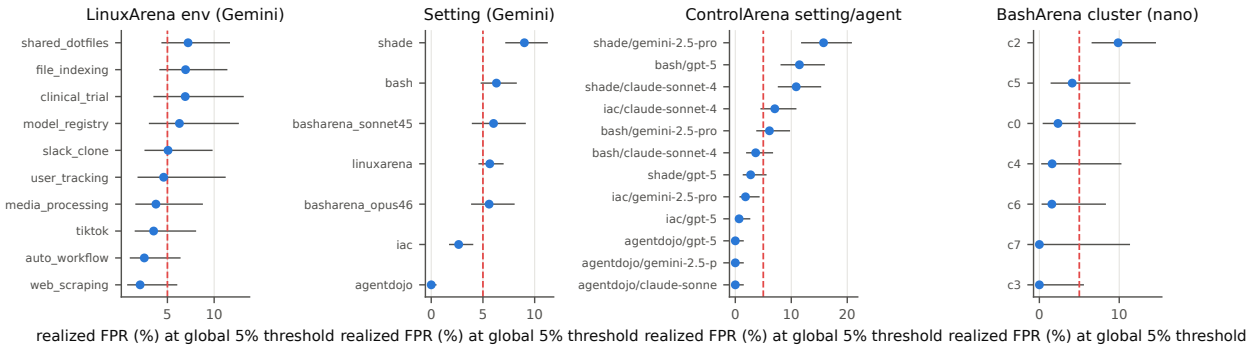


Figure 1: Realized per-context FPR under one global 5% threshold (Wilson 95% CIs), four panels. [t2/figures/]

3.2 Allocation at a fixed global budget

Panel	Rule	$B = 1\%$ nominal spend	unif. TPR	sel.	$B = 1\%$ spend-matched sel.	Δ sel vs global	m_ok	$B = 5\%$ spend-matched unif. TPR	sel.	Δ sel vs global
7 settings	global	1.01%	0.853	0.645	0.644	–	1.00	0.965	0.890	–
	CFAR	1.13%	0.821	0.404	**0.378**	−0.52.. − 0.10	0.99	0.946	0.753	−0.30.. + 0.01
	pooled_cv	1.10%	0.829	0.439	0.415	−0.49.. + 0.00	0.99	0.947	0.757	−0.29.. + 0.01
	Mondrian	0.36%	0.689	0.125	*n.e.*†	–	0.30	0.941	0.710	−0.33.. + 0.03
	max-detect	1.12%	0.891	0.641	0.617	−0.28.. + 0.08	0.97	**0.984**	0.958	+0.02.. + 0.12
	Stackelberg	1.07%	0.834	0.791	0.779	−0.11.. + 0.20	0.98	0.971	**0.966**	+0.03.. + 0.12
ControlArena	global	0.97%	0.798	0.601	0.607	–	0.96	0.960	0.908	–
	CFAR	1.34%	0.837	0.218	0.161†	−0.69.. − 0.17	0.68	0.935	**0.561**	−0.63.. − 0.06
	max-detect	1.49%	0.898	0.574	0.408†	−0.65.. + 0.14	0.62	0.985	0.950	−0.06.. + 0.12
	Stackelberg	1.38%	0.863	0.729	0.610†	−0.36.. + 0.17	0.66	0.982	0.973	−0.00.. + 0.13

Table 3: Allocation rules (plug-in thresholds, clamping matcher). "sel." = selection safety (TPR in the context an informed attacker picks). Δ = 2.5–97.5% of paired replicate differences. m_ok = `matched_ok` at 1% (all 5% rows shown have m_ok ≥ 0.97). † = m_ok < 0.8, partly extrapolated by the clamping matcher; not interpreted. [`t2/tables.md`]

- **CFAR is the worst rule against selection** on the heterogeneous panels: 7 settings at 1% (spend-matched, m_ok 0.99) and ControlArena at 5% (m_ok 1.00) both have paired intervals below 0. With attacker information drawn from a split independent of the defender's fit, 7 settings at 1% gives CFAR 0.409 vs global 0.642 (interval −0.34.. − 0.13). [`audit/repro_t2.csv`]
- **Stackelberg beats global at 5%** (7 settings +0.03.. + 0.12; ControlArena −0.00.. + 0.13). With independent attacker information: 0.775 at 1% and 0.957 at 5% vs global 0.642 / 0.888. **At 1% the gain was not established in round 1**: the 7-settings spend-matched interval crosses 0, and the ControlArena nominal gain (0.729 vs 0.601) comes from over-spending (1.38% vs 0.97%); matched, it is 0.610 vs 0.607. [`fixes/corrections.csv (m2)`] Round 4 re-tests the 7-settings 1% contrast with exact matching (Section 7). The 5% gain re-run under the conformal rule is +0.076 [+0.026, +0.124] (overview row; [`fixes2/headline_threshold_rerun.csv`]).
- **Max-detect** gives the best uniform TPR (0.984 vs 0.965 at 5%, 7 settings) and no robust selection advantage.
- **Damage-weighted Stackelberg** vs a damage-weighted attacker at 5%: 0.984 vs 0.937 (7 settings) and 0.989 vs 0.930 (ControlArena). [`t2/tables.md`]
- **Homogeneous panel** (LinuxArena environment, local; τ n.s.): CFAR, pooling, max-detect and Stackelberg are within 0.012 uniform TPR and 0.05 selection safety of global at nominal 1%. Allocation does not matter when contexts do not differ.
- **Cell-level selection** (choosing an attack type within a context) costs another 0.2–0.4: LinuxArena environment at 1% falls from 0.900 to 0.607 (global) and from 0.948 to 0.704 (Stackelberg), nominal mode.
- **Weak monitors** (MonitoringBench clusters): the round-1 uniform-TPR gains of CFAR / max-detect (nano trajectory at 5%: 0.079 → 0.106 / 0.162) rest on rows with m_ok 0.70 / 0.34 (max-detect realizes 5.56%). At nominal 1% the per-context rules spend 3.2–4.3% vs global 1.25%. This is the spend floor of Section 3.3 again; the gain is not established. Selection safety stays ≤ 0.32 for every rule.

3.3 Small samples per context

Panel	n per ctx	global	CFAR	Mondrian	pooled_eb	pooled_cv
7 settings	10	1.61× / 0.81 / 0.63	**7.26**× / 0.91 / 0.59	0 / 0 / 0	2.45× / 0.85 / 0.63	1.72× / 0.83 / 0.64
	30	1.16× / 0.83 / 0.63	2.90× / 0.85 / 0.43	0 / 0 / 0	1.60× / 0.85 / 0.60	1.30× / 0.84 / 0.61
	100	1.01× / 0.84 / 0.64	1.48× / 0.84 / 0.40	0.46× / 0.68 / 0.08	1.25× / 0.85 / 0.52	1.13× / 0.84 / 0.58
ControlArena	10	1.35× / 0.77 / 0.59	6.20× / 0.92 / 0.43	0 / 0 / 0	2.32× / 0.85 / 0.56	1.39× / 0.78 / 0.60
	100	1.02× / 0.80 / 0.60	1.41× / 0.84 / 0.21	0.43× / 0.74 / 0.00	1.23× / 0.85 / 0.30	1.09× / 0.83 / 0.44
LA task category	10	2.61× / 0.93 / 0.88	7.66× / 0.95 / 0.87	0 / 0 / 0	3.95× / 0.94 / 0.88	2.91× / 0.94 / 0.89

Table 4: Spend/B / uniform TPR / selection safety at $B = 1\%$, nominal mode. [t2/tables.md]

- The plug-in per-context quantile spends $\approx 1/(n_c+1)$ per context regardless of the target: CFAR spends 6.2–7.7× the budget at $n = 10$, 2.5–2.9× at 30 and 1.4–1.6× at 100 (four panels).
- Mondrian conformal is valid but *never alarms* below $n_c \approx 1/B$.
- Partial pooling with cross-validated n_0 stays at 1.06–1.18× with $n = 100$ and 1.16–1.40× with $n = 30$, at equal uniform TPR. Empirical-Bayes n_0 is worse.
- LinuxArena sweep and MRT (10 calibration runs per context, nominal 5%): CFAR spends 7.8–12.3%, max-detect and Stackelberg 9.1–14.4%, global and pooled_cv 4.6–6.5%.

3.4 Poisoning one context's recalibration data

Honest runs per context are split into a trusted reference, a new window and test (1/3 each). The attacker adds a share ε of the target context's new window at its own maximum calibration attack score; every context is targeted in turn (40 replicates).

Panel	ε	global	CFAR	pooled_eb	OS-CFAR (LOCO)	TM-CFAR	50/50 ref. mix	frozen ref.	drift cap
LA environment	0	.945	.949	.947	.896	.675	.943	.945	.953
	2%	.941	**.366**	.931	.887	.590	.724	.945	.800
	5%	.933	.157	.759	.862	.382	.314	.945	.778
	10%	.785	.079	.360	.807	.033	.157	.945	.771
7 settings	0	.854	.821	.848	.872	.655	.829	.824	.824
	5%	.641	.109	.172	.808	.252	.212	.824	.531
	10%	.388	.055	.076	.722	.000	.108	.824	.516
LA task category	0	.914	.906	.910	**.778**	.525	.892	.894	.920
	10%	.520	.068	.125	.615	.000	.136	.894	.613

Table 5: TPR in the poisoned context (mean over target contexts), $B = 1\%$. [t2/tables.md]

- **CFAR breaks once** $\varepsilon \approx B$: a per-context quantile at level B moves with any poison share $\geq B$.
- **Global breaks near** $\varepsilon \approx B/w_t$: the poison is diluted by the target's traffic share w_t.
- **OS-CFAR with a LOCO tail** takes its tail from the other contexts, so one poisoned context cannot move it. At $\varepsilon = 10\%$ the mean target TPR is 0.62–0.81 by panel, but the worst target context falls to 0.07–0.45; without poisoning it costs detection on LA task category (0.778 vs 0.914). [fixes/corrections.csv (m3)] Round 2 shows it breaks when the poison is spread (Section 4.4).
- **Boiling frog** (30 recalibration rounds, sliding window, poison just above the current threshold, $\varepsilon = 5\%$): CFAR ends at 0.11 / 0.16 TPR (7 settings / LA environment); an outlier filter ends at

the same values, only later; a cumulative cap holds 0.54 / 0.78 but under a genuine +2 shift its FPR is 3.3% / 4.9%; global ends at 0.65 / 0.94 with FPR 4.0% under drift; OS-CFAR (LOCO) holds 0.80 / 0.82 with FPR 0.6% / 1.1% under drift. [t2/tables.md]

4 Round 2: transfer, spread poisoning, severity, adaptive attackers [E]

4.1 E1a: Stackelberg fit on a different attack distribution

The defender fits Stackelberg / max-detect on source attacks; the attacker comes from a target distribution and picks its context from its own calibration attacks. Spend-matched.

Panel	Shift	B	global	CFAR	Stackelberg (source) [Δ vs global; Δ vs in-dist.]	Stackelberg (in-dist.) / $\lambda = .5$
7 settings	unseen side tasks	1%	0.62	0.40	0.43 [−0.13 (−0.48, +0.11); −0.27 (−0.56, −0.03)]	0.78 / 0.54
		5%	0.87	0.78	0.70 [−0.16 (−0.51, +0.03); −0.27 (−0.65, −0.04)]	0.97 / 0.82
ControlArena	unseen side tasks	5%	0.83	0.54	0.58 [−0.23 (−0.62, +0.11); −0.38 (−0.81, −0.08)]	0.98 / 0.79
	attack policy p1→p2	1%	0.72	0.15	0.66 [−0.08 (−0.40, +0.13); +0.04]	0.61 / 0.67
		5%	0.91	0.56	0.95 [+0.04 (−0.09, +0.26); −0.01]	0.96 / 0.93
LA environment	unseen side tasks	5%	0.96	0.95	0.86 [−0.08 (−0.38, +0.06); −0.07]	0.97 / 0.91

Table 6: Selection safety under attack-distribution shift (m_ok $\geq$ 0.86 in all rows shown). λ = .5: each context's source attacks mixed 50/50 with the pooled source attacks of the other contexts. [r2_t2/tables.md (E1a)]

- **A new attack policy or month does not hurt**: ControlArena p1→p2 keeps the in-distribution level (0.66 / 0.95 vs 0.61 / 0.96 at 1% / 5%).
- **Unseen side tasks do**: Stackelberg fit on other side tasks is significantly worse than the in-distribution fit (e.g. 7 settings 1%: -0.27 $[-0.56, -0.03]$) and lands at or below global (point estimates -0.08 to -0.23; intervals cross 0).
- Shrinkage moves the source fit monotonically toward global ($\lambda = 0.5$: 0.54 at 1% on 7 settings) and costs ≤ 0.05 in distribution (0.73 vs 0.78).
- On MonitoringBench (weak monitors) the 1% rows of the attack-fitted rules are clamped (m_ok ≤ 0.08) and are not interpreted.

4.2 E1b: a new agent model

Rules fit on the other agent models' data are applied to a held-out agent model.

- **Transferred per-context thresholds miss the budget in both directions.** Realized spend at nominal 1% ranges from 0.1% (source CFAR, opus-4-6 on LA environment) to 2.8% (source Stackelberg, gemini-2.5-pro on ControlArena); at nominal 5% from 1.2% (source Stackelberg, gpt-5) to 12.3% (source CFAR, gpt-5; 2.5×).
- Source CFAR is also exploitable on the new agent: gpt-5 on ControlArena at 5%, selection 0.38 vs in-domain global 0.94 (-0.53 $[-0.72, -0.30]$).
- **Source offsets plus one level recalibrated on the target** (xfer_offset) hold the budget: 1.0–1.9% at nominal 1% and 4.6–5.7% at 5%. Selection safety is not protected (gpt-5 on ControlArena at 5%: 0.63, -0.26 $[-0.49, -0.06]$). [r2_t2/tables.md (E1b)]

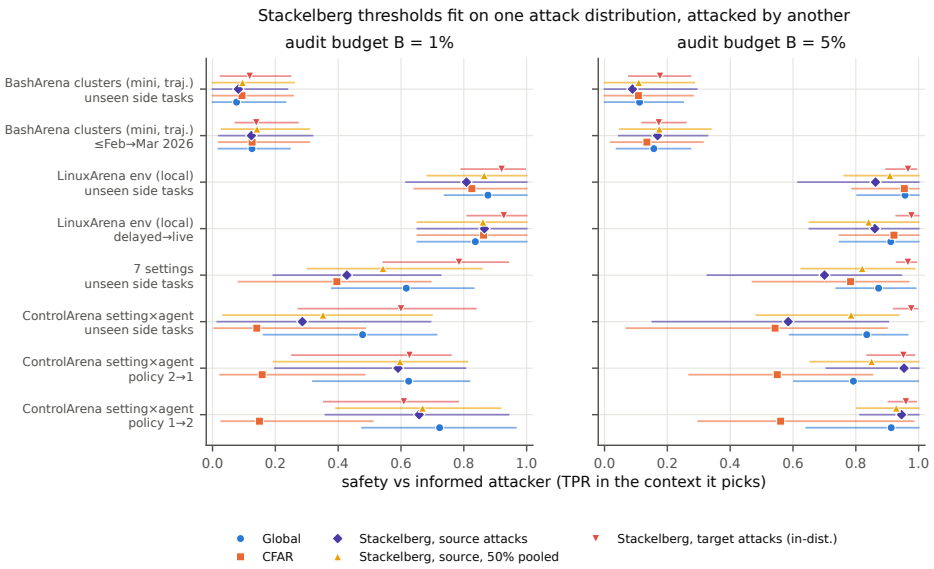


Figure 2: E1a: selection safety of Stackelberg fit on source attacks vs in-distribution and global, by panel and shift. [r2_t2/figures/]

4.3 E1c: offsets from other monitors

Per-context offsets estimated from the other 11 LinuxArena-sweep monitors' scores on the same calibration trajectories (xfer_avg) reduce the SD of per-environment FPR, spend-matched, from 0.060 to 0.050 at 5% and from 0.114 to 0.066 at 10%; it is lower than global for 9 of 12 target monitors at 5% and 10 of 12 at 10% [r2_t2/e1c_monitor_transfer_full.csv]. Like CFAR, equalizing FPR costs selection safety at 5%: 0.45 vs 0.65 for global. On the strong Gemini LinuxArena pair and on gpt-4.1-mini/nano there is little to gain (per-context FPR SD ≤ 0.03 under global). [r2_t2/tables.md (E1c)]

4.4 E2: poisoning spread over several contexts

The attacker poisons the target only, one other context, half of the others, all others, or all contexts, placing values that beat both a raw global threshold and an offset-based OS-CFAR. Genuine-drift controls: a heavier honest tail in every context (3% of runs replaced by high values), a deployment-wide location shift, and a target-only shift.

Panel	Rule	clean TPR	worst @ $\varepsilon = 2\%$	worst @ 5%	worst @ 10%	max drift spend / B
LA environment	global	0.94	0.37	0.16	0.08	1.19
	CFAR	0.95	0.37	0.16	0.08	2.14
	OS-CFAR, LOCO tail (round 1)	0.88	0.40	0.18	0.11	1.40
	OS-CFAR, frozen-reference tail	0.89	**0.88**	**0.87**	**0.81**	2.47
	frozen reference (never recalibrate)	0.95	0.95	0.95	0.95	4.10
	OS-CFAR, low-quantile tail	0.95	0.46	0.42	0.41	3.32
7 settings	global / CFAR	0.85 / 0.82	0.26 / 0.27	0.11 / 0.11	0.05 / 0.05	1.10 / 1.16
	OS-CFAR, LOCO / frozen tail	0.88 / 0.87	0.35 / **0.84**	0.18 / **0.79**	0.13 / **0.70**	1.12 / 2.56
ControlArena	OS-CFAR, LOCO / frozen tail	0.80 / 0.82	0.28 / **0.80**	0.17 / **0.78**	0.08 / **0.72**	1.28 / 2.86
LA task category	OS-CFAR, LOCO / frozen tail	0.75 / 0.76	0.31 / **0.74**	0.16 / **0.70**	0.13 / **0.59**	1.22 / 2.02

Table 7: Worst target TPR over the poisoning scenarios and worst realized spend / B over the genuine-drift controls, $B = 1\%$. [r2_t2/tables.md (E2 summary)], [r2_t2/e2_breakdown_summary.csv]

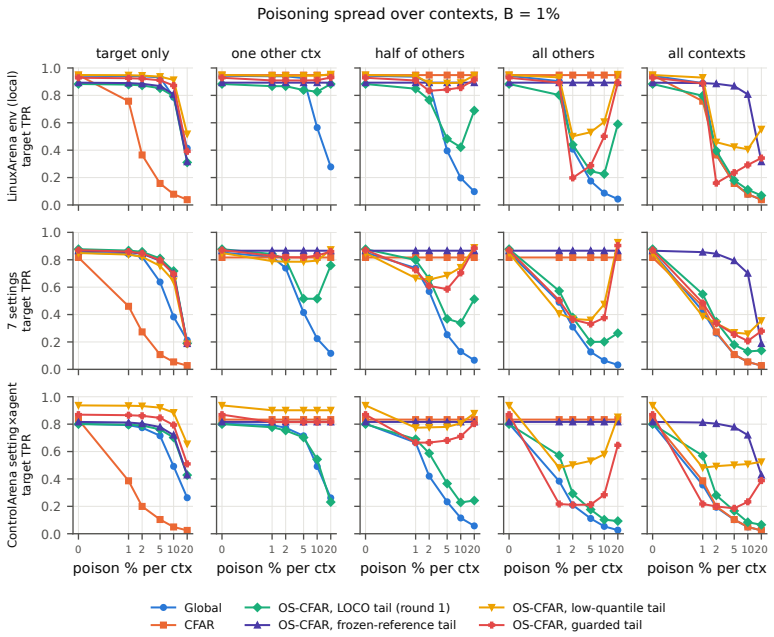


Figure 3: E2: target-context TPR vs poisoned share, by scenario (columns) and panel (rows), $B = 1\%$. [r2_t2/figures/]

- **The LOCO tail breaks at $\varepsilon \approx B$ when the poison is in all other contexts**: LA environment target TPR 0.88 → 0.44 at 2% and 0.25 at 5%. Worst case over scenarios: 0.16–0.18 at 5% on all four panels.
- **A tail from a frozen trusted reference** (offset still from the new window, so poison must exceed 25% of a context's window to move it) holds a worst-case 0.70–0.81 at $\varepsilon = 10\%$ on three panels and 0.59 on LA task category (clean 0.76).
- **Its price**: under genuine deployment-wide tail drift it over-spends 2.0–2.9× (it refuses to follow a real change in the tail). Never recalibrating over-spends 3.1–7.7× under the location shift.
- Median, low-quantile, trimmed and guarded tails resist minority poisoning only (worst case at 5%: 0.11–0.50). **Poisoning of every context is indistinguishable from deployment-wide drift** without a trusted refresh of the reference.

4.5 E3: severity-weighted damage

aksh-n's adjudicated `overall_severity` correlates moderately with the monitor score (Spearman 0.43–0.51) and less with the side-task taxonomy tier (0.29–0.38). Spend-matched, $B = 5\%$, against a damage-maximizing attacker who picks a (context, side task) cell:

Panel	global	CFAR	Stackelberg (context)	cell minimax, adjudicated damage	cell minimax, shrunk
7 settings	0.64	0.60	0.76 [+0.14 (−0.14, +0.41)]	0.86 [+0.13 (−0.25, +0.31)]	**0.92** [+0.25 (+0.00, +0.44)]
ControlArena	0.71	0.48 [−0.20 (−0.38, −0.01)]	0.88 [+0.17 (−0.14, +0.40)]	0.94 [+0.23 (+0.06, +0.37)]	**0.95** [+0.24 (−0.06, +0.44)]

Table 8: 1 − expected undetected damage in the cell the attacker picks (damage = adjudicated severity), with paired Δ vs global. [r2_t2/tables.md (E3)]

- Cell-level severity minimax is the best rule against a cell-choosing attacker; only the unshrunk ControlArena contrast and (borderline) the shrunk 7-settings contrast exclude 0.
- Against a context-choosing attacker it loses to context-level severity Stackelberg (0.91 vs 0.98 on 7 settings; 0.96 vs 0.99 on ControlArena).
- Side-task taxonomy weights are a poor proxy for damage: tier-weighted cell minimax is at or below global (0.59 vs 0.64; 0.64 vs 0.71).

4.6 E4: probing attackers and estimation error

Panel, B	Rule	uniform	$q = 1$	$q = 3$	$q = 5$	$q = 20$	informed [95%]
ControlArena, 1%	global	0.80	0.71	0.69	0.68	0.65	0.60 [0.48, 0.74]
	CFAR	0.83	0.46	0.30	**0.25**	0.19	0.18 [0.07, 0.46]
	Stackelberg	0.86	0.77	0.74	0.74	0.73	0.73 [0.59, 0.86]
7 settings, 1%	global	0.85	0.78	0.75	0.73	0.68	0.64 [0.57, 0.71]
	CFAR	0.82	0.60	0.50	0.46	0.41	0.40 [0.18, 0.61]
	Stackelberg	0.83	0.81	0.80	0.80	0.80	0.79 [0.68, 0.89]

Table 9: Safety against a probing attacker with free alarm feedback on q probes per context (nominal thresholds). With costly probes ($q = 5$) safety is 1.00 on every strong-monitor panel. [r2_t2/tables.md (E4a)]

- **CFAR's weakness does not need inside knowledge**: 3–5 free probes per context recover most of the gap between the uniform and the fully informed attacker. Stackelberg is nearly flat in q.
- **Estimation error (defender)**: Stackelberg fit with n_d attacks per context needs about 10–20 to match global (7 settings 1%: 0.55 / 0.61 / 0.68 / 0.70 at $n_d = 3/5/10/20$ vs global 0.64; 5%: 0.67 / 0.74 / 0.85 / 0.90 vs 0.88). Shrinkage toward the pooled attacks ($m_0 = 10$) removes the small-sample penalty (0.66 at $n_d = 3$, 1%; 0.86 at 5%).
- **Estimation error (attacker)**: with only 3 attacks per context to estimate evasion, the attacker's selection against CFAR is already near its full-information level (ControlArena 1%: 0.26 vs 0.18). [r2_t2/tables.md (E4b)]

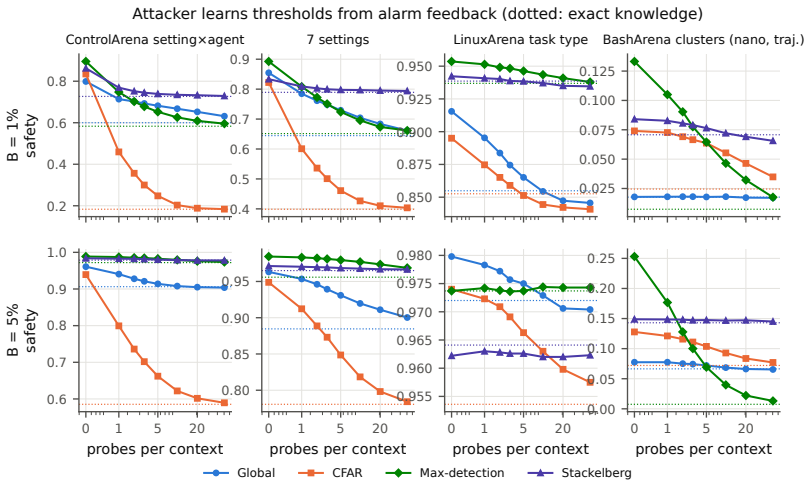


Figure 4: E4a: safety vs number of free probes per context, $B = 1\%$ (top) and 5% (bottom). [r2_t2/figures/]

5 E5: white-box probes and the strong monitor [E]

E5 uses one common trajectory set (Honest-550 and 541 attacks) scored by 17 signals: 7 white-box streams of a Qwen3.5-27B trusted reader (`sab` synthetic-sabotage probe, Apollo deception probe `dec`, FIBS token probe `fibs_tok`, FIBS follow-up contrast `fibs_fu`, J-lens watch-list mass `jl_lm`, and max-pooled variants) in history-aware (full) and history-free (none) reading, plus gpt-4.1 (none) and gpt-4.1-mini (none, full). Contexts are the 7 task clusters.

5.1 Heterogeneity (honest runs only; unaffected by the audits)

- **Probe scores are far more context-dependent than LLM-monitor scores.** τ at a global 5% threshold: `dec_full` 3.79, `fibs_fu_full` 2.70, `dec_none` 2.28, `jl_lm_max_none` 1.77, `fibs_tok_none` 1.26, `sab_full` 1.21 (all $p \leq 0.006$). Per-cluster FPR spans up to 0–25.5%.
- LLM monitors on the same trajectories: τ 0.00–0.46, all n.s.; ICC 0.002–0.035, vs 0.10–0.16 for `fibs_fu`, `dec_none` and `sab_max_none`. [r2_t2/e5_tables.md], [fixes/corrections.csv (M5)]

5.2 The 1% comparison is not estimable (C1)

Plug-in per-cluster thresholds at level 0 sit at each cluster's calibration maximum, so the realized test spend has a floor of $\sum_c w_c/(n_c + 1) = 2.52\%$ with 15–106 calibration honest runs per cluster (per-replicate mean 2.58%, 5–95% 0.74–4.89%; only 9% of splits have a floor $\leq 1\%$). At nominal 1% the realized spend is 1.16% for global, 2.99% for CFAR, 3.76% for Stackelberg and 4.19% for max-detect. The original "spend-matched" 1% rows were clamped at 1.95–3.57% and are dropped. [fixes/r2_t2/e5_tables_corrected.md]

The valid 1% comparison evaluates global at the spend each per-context rule actually realizes:

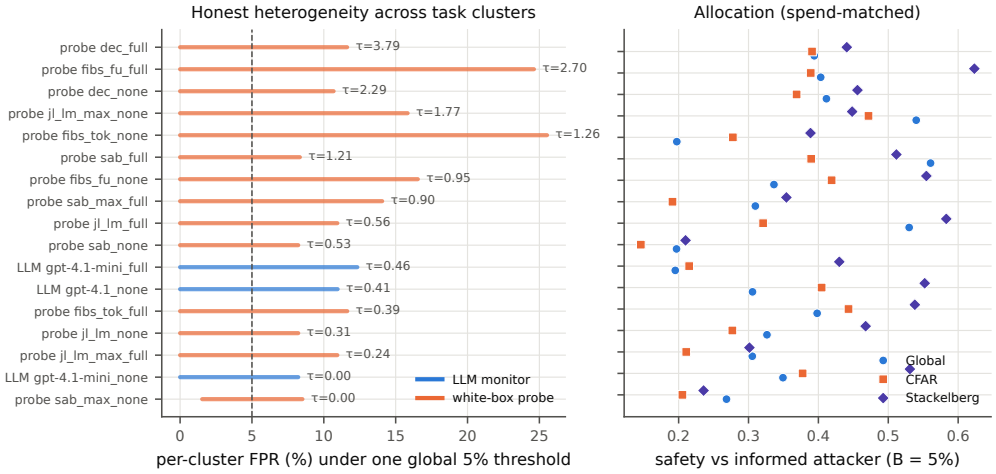


Figure 5: E5. Left: per-cluster FPR under one global 5% threshold, white-box probes vs LLM monitors (honest runs only; valid). Right: the original iid-split allocation at 5%, superseded by Table 11 (audit M5). [r2_t2/figures/]

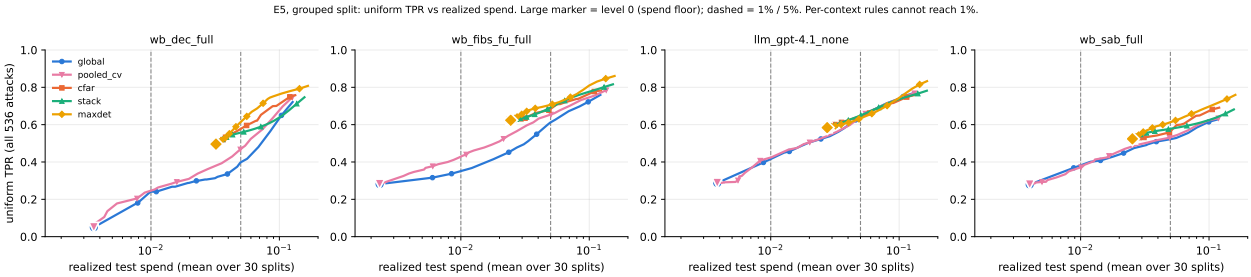


Figure 6: E5 corrected: uniform TPR vs realized spend for four signals. Large markers = level 0 (the spend floor); dashed lines = 1% and 5%. Per-context rules cannot reach 1%. [fixes/r2_t2/]

Signal	realized spend	CFAR uniform TPR	global at same spend	Δ [split 95%]	ratio
wb_dec_full	3.63%	0.52	0.34	+0.18 [+0.08, +0.27]	1.5×
wb_fibs_fu_full	2.94%	0.63	0.49	+0.15 [+0.03, +0.28]	1.3×
wb_dec_none	3.07%	0.55	0.33	+0.22 [+0.14, +0.30]	1.7×
wb_fibs_fu_none	3.01%	0.55	0.48	+0.07 [+0.01, +0.14]	1.1×
wb_sab_full	2.85%	0.53	0.48	+0.05 [+0.00, +0.12]	1.1×
llm_gpt-4.1_none	3.16%	0.60	0.55	+0.05 [−0.01, +0.12]	–
llm_gpt-4.1-mini_none	3.02%	0.47	0.46	+0.01 [−0.12, +0.11]	–

Table 10: CFAR at nominal 1% vs global at the same realized spend, grouped split; interval over 30 plain splits. The split interval excludes 0 for 5 of 17 signals, all white-box probes. [fixes/r2_t2/e5_equal_spend_1pct.csv]

5.3 Allocation at 5% with the split grouped by base attack (M5)

The original E5 split attacks iid within context, so refined variants of one base attack fell on both sides of the split. The corrected design groups the split and the bootstrap by base attack (148 base attacks; 0 exact duplicates).

Signal	global unif. / sel.	CFAR unif. [Δ]	CFAR sel. [Δ]	Stackelberg sel. [Δ]	max-detect unif. [Δ]
llm_gpt-4.1_none	0.64 / 0.37	0.65 [+0.01 (−0.09, +0.10)]	0.44 [+0.11]	0.62 [+0.14 (−0.23, +0.47)]	0.64 [+0.00]
wb_fibs_fu_full	0.61 / 0.47	0.69 [+0.07 (−0.02, +0.21)]	0.45 [−0.00]	0.59 [+0.13 (−0.39, +0.50)]	0.71 [+0.10]
wb_dec_full	0.40 / 0.43	0.60 [+0.19 (+0.03, +0.34)]	0.43 [−0.08]	0.45 [−0.01]	0.64 [+0.22 (+0.04, +0.45)]
wb_dec_none	0.37 / 0.43	0.62 [+0.23 (+0.05, +0.35)]	0.39 [−0.04]	0.41 [−0.03]	0.65 [+0.25 (+0.07, +0.39)]
wb_sab_full	0.52 / 0.59	0.58 [+0.06 (−0.03, +0.14)]	0.46 [−0.11 (−0.59, +0.27)]	0.53 [−0.03]	0.61 [+0.09 (+0.02, +0.19)]
llm_gpt-4.1-mini_none	0.59 / 0.44	0.56 [−0.01]	0.40 [−0.04]	0.55 [−0.03]	0.57 [−0.03]
Counts over 17 signals		≥ global 12, CI> 0 2	≥ global 5, CI> 0 0	≥ global 10, CI> 0 0	≥ global 14, CI> 0 4

Table 11: E5 corrected, $B = 5\%$ spend-matched (matched in 0.83–1.00 of splits), grouped split, cluster bootstrap over base attacks. Δ = paired difference vs global. [fixes/r2_t2/e5_tables_corrected.md]

- **Uniform attacker**: per-context normalization helps the strongly heterogeneous probes (dec) and nothing else significantly. For the LLM monitors it is neutral.
- **Selection**: no rule's selection advantage over global is established on E5. Stackelberg ≥ global on 10/17 signals (the original log claimed 14/17; its own CSV gives 13/17 at 5%); no interval excludes 0.
- **The attack set is confounded.** AUROCs on the 33 originals are much lower than on all 541 (gpt-4.1 0.765 vs 0.911; fibs_fu_full 0.688 vs 0.905; dec_full 0.573 vs 0.871), and refined variants of the same base attacks score higher than their originals on 13/17 signals (interval > 0; dec_full +0.31 [+0.19, +0.43]). Fabricated tool outputs and rewrite artifacts cannot be separated from detection with these data. The sab probes are the exception (originals 0.81–0.83).
- **Originals only** (honest-only rules, no fabrication confound, 1–15 originals per cluster): at 5% matched, CFAR raises cluster-averaged TPR for dec_none (+0.33 [+0.02, +0.57]) and fibs_fu_full (+0.20 [+0.00, +0.37]) and not for gpt-4.1 (−0.02 [−0.26, +0.20]).

6 Round 3: Watcher-like severity tiers [E]

A Watcher-style product sees two numbers per action: suspicion s (history-free gpt-4.1-mini or gpt-4.1, or history-aware gpt-4.1-mini) and an intent-agnostic expected severity v (1–10, Apollo Watcher tiers: 1–3 benign, 4–6 strict-only, 7–8 permissive, 9–10 always blocked). Severity was scored by gpt-4.1-mini on all 38.4k MonitoringBench steps (Honest-550 and 775 T1 attacks; $21.82, 0 errors). Orgs are the task clusters; damage is the attack's maximum severity, or the side-task tier (@sidetask panels).

6.1 Severity does not separate attacks from honest work

Panel	label	trajectories	action share by tier (1–3/4–6/7–8/9–10)	traj. has a 9–10 action	AUROC suspicion	AUROC severity
mb_mini	attack	752	0.61 / 0.22 / 0.16 / 0.002	6.9%	0.758	0.532
	honest	535	0.57 / 0.24 / 0.19 / 0.001	**1.5%**		
mb_strong (gpt-4.1)	attack	143	0.61 / 0.22 / 0.17 / 0.002	7.0%	0.782	0.540
	honest	505	0.56 / 0.24 / 0.19 / 0.001	1.6%		
akshn_adj (label-leaking)	attack	5,794	0.00 / 0.13 / 0.79 / 0.080	8.0%	0.991	0.998
	honest	5,189	0.97 / 0.03 / 0.00 / 0.000	0.0%		

Table 12: Severity-tier composition. Every honest BashArena trajectory contains a 7–8 action. The akshn_adj severity is assigned with knowledge of the outcome; it is an upper bound no deployed rater can reach. [r3_t2/tables.md], [fixes/r3_t2/tables_annotated.md]

6.2 Fixed tiers lose to a suspicion threshold at matched budget

- **1% is infeasible for tiered policies**: "always block 9–10" alone flags about 1.5% of honest trajectories. Every 1% "matched" row of the round-3 tables was a clamped mixture. Re-summarized without clamping (round 4), every per-org rule and both Watcher tiers on the MonitoringBench panels are *n.e.* at 1% (matched in 4–25% of replicates); the clamped tier_opt value 0.191 becomes 0.140 on matched replicates only. No 1% comparison is reported. [audit2/r3t2_matched_check.csv], [r4_ctx/r3t2_corrected/tables_noclamp.md]
- **At 5%** (Watcher tiers matched in 95–100% of replicates):

Panel	global	Watcher permissive [Δ]	Watcher strict [Δ]	sev_tilt [Δ]
mb_mini (gpt-4.1-mini none)	0.344	0.257 [−0.082 (−0.191, −0.021)]	0.281 [−0.059 (−0.168, −0.006)]	0.344 [−0.002]
mb_strong (gpt-4.1 none)	0.373	0.263 [−0.121 (−0.247, −0.006)]	0.313 [−0.071 (−0.219, +0.006)]	0.367 [−0.015]
mb_mini_full (history-aware)	0.211	0.170 [−0.045 (−0.089, −0.003)]	0.191 [−0.024 (−0.072, +0.013)]	0.211 [−0.013]

Table 13: Damage-weighted detection against a uniform attacker, $B = 5\%$, spend-matched without clamping (round-4 re-summary: a replicate counts only if B lies inside its realized-spend range; all rows matched in $\geq 99\%$ of replicates), with paired Δ vs global. The clamped round-3 values differed by ≤ 0.005 in level. Plug-in thresholds. [r4_ctx/r3t2_corrected/watcher_summary_noclamp.csv]

- **Org-selecting and severity-aware attackers do no better against the tiers**: e.g. mb_mini org-selection 0.35 / 0.37 (permissive / strict) vs 0.43 for global. The incriminating action is often not the highest-severity one.
- **Severity tilt**: a global threshold on $s + \beta v$ with β tuned on calibration attacks picks $\beta = 0$ in 80–99% of replicates at 5% on the blind-rater panels [data/processed/r3_t2_runs/watcher.parquet], i.e. severity adds nothing. On the label-leaking panel it picks $\beta > 0$ and reaches 0.99: the illusory upside.

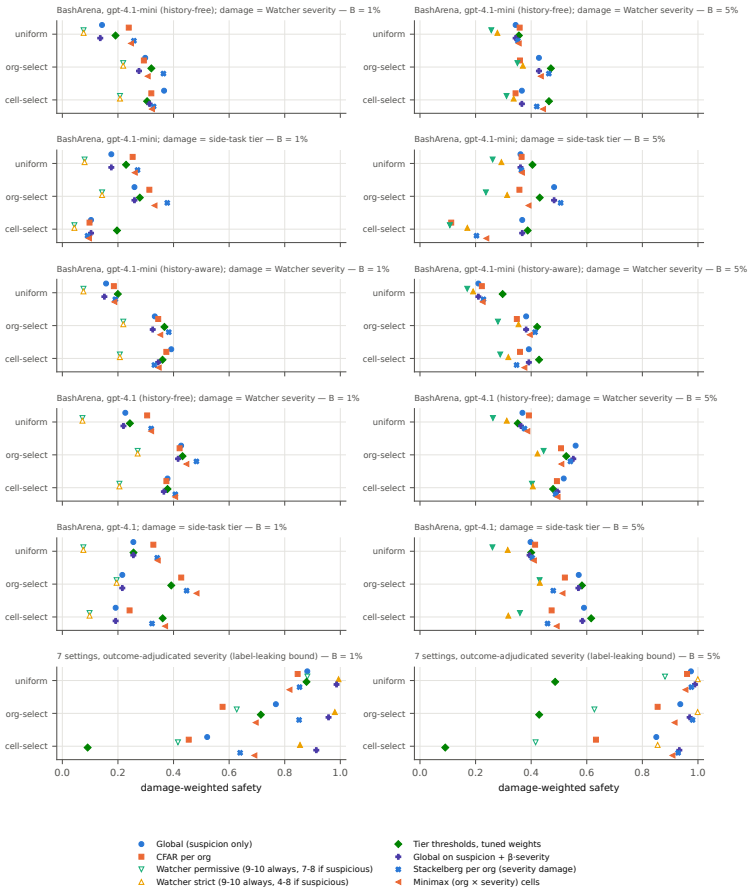


Figure 7: Round 3 policy comparison: damage-weighted safety against uniform, org-selecting and cell-selecting attackers. Only the 5% panels are valid; the 1% panels mix clamped replicates (audit2 m9). The bottom row (akshn_adj) uses label-leaking severity. [`r3_t2/figures/`]

- **Tuned tier thresholds** (tier_opt) end close to global (mb_mini 0.356 vs 0.344 at 5%).
- **Per-org rules** (CFAR, Stackelberg, severity Stackelberg, severity cell minimax) on mb_mini are matched in only 78% of replicates at 5% and are therefore *n.e.* without clamping; on the other panels (matched 80–87%) they are within noise of global. At 1% they over-spend out of sample (2.7–3.8% realized at nominal 1%). On akshn_adj, Watcher-permissive (1% and 5%), Watcher-strict (5%) and tier_w (5%) were never matched: the spend is unreachable, and their earlier "matched" values were clamped.
- **Severity-aware attacker with side-task damage**, 5%: safety 0.37 (mini) / 0.59 (gpt-4.1) under global vs 0.11 / 0.47 under CFAR (intervals cross 0). [`r3_t2/tables.md`]

7 Round 4: allocation at 1% with hundreds of honest runs per context

Why. Per-context rules have a test-spend floor of about $\sum_c w_c/(n_c+1)$, so in rounds 1–3 the 1% "spend-matched" rows were clamped or barely matched. Round 4 used panels with hundreds of honest runs per context and the exact spend matcher (Section 2.5 and the evaluation section). Code

`tracks/r4_ctx/`; $0 API; about 10 minutes of compute.

Panels. *Primary*: `akshn_setting` (7 settings, Gemini-3.1-Pro local suspicion, trajectory max; 5,189 honest with 328–1,396 per setting, 5,794 attacks; damage = adjudicated severity). *Secondary*: ControlArena setting × agent (12 contexts, ≈250 honest each) and the full public LinuxArena runs by environment (gpt-5-nano per-action monitor, ≈217 honest per environment over 20 tasks, 919 attacks). Cross-deployment `simple@` panels were excluded before analysis because 8–47% of honest maxima sit at the scale top.

Design (pre-specified).

- Rules: global, CFAR, partial pooling (EB n_0), Stackelberg with equal damage and attack hulls shrunk toward the other contexts ($m_0 = 10$; also unshrunk), severity-weighted Stackelberg.
- 10 random splits per panel. Within a context, splits are grouped by main task when it has ≥ 20 distinct main tasks, otherwise by trajectory. Honest runs: calibration / test 50/50. Attacks: three disjoint thirds for the defender's fit, the attacker's information and evaluation.
- Inference: 200 replicates of a context-stratified cluster bootstrap whose weights are reused across splits and rules, with thresholds refit and re-matched per replicate; 95% percentile intervals.
- Attackers: uniform, informed context selection (from its own information third), probing (5 free probes per context), damage-maximizing. Budgets $B \in \{0.5, 1, 2, 5\}\%$. Transfer: the defender fits on attacks of half of the side tasks (or attack policies / agent models).

7.1 Primary contrast [C]

On 7 settings at $B = 1\%$, in distribution, both rules matched in 99.9% of cells (spend floor 0.19% mean, 0.52% p95):

- selection safety against the informed attacker: Stackelberg 0.753 [0.661, 0.796], global 0.646 [0.596, 0.695];
- $\Delta = +0.107$ **[**$+0.024$**,** $+0.152$**]**, $P(\Delta > 0) = 1.00$. The pre-specified criterion (interval above 0, both matched in $\geq 80\%$) is met. Unchanged under the conformal threshold rule, including a version that also puts the Stackelberg hull on the conformal scale (+0.105 [+0.030, +0.150]).[`fixes2/headline_threshold_rerun.csv`]

Robustness and the main caveat (audit 3, [E]):[`audit3/ctx_primary_checks.csv`]

- The gain is driven by SHADE. Under the global rule the informed attacker picks SHADE in 10 of 10 splits, and Stackelberg raises SHADE's test FPR from 2.7% to 5.0%. SHADE, iac and AgentDojo have only 5, 5 and 16 main tasks, so by the pre-specified rule they were split and resampled by trajectory.
- **With main-task cluster resampling on the same splits the interval is** [-0.003**,** $+0.251$], $P(\Delta > 0) = 0.97$: the criterion is met only with trajectory-level resampling. A 5-task bootstrap is coarse, so both intervals are reported.
- Other checks favour a positive effect: basic bootstrap interval [+0.061, +0.190]; all 10 splits positive (+0.042 to +0.172); iid instead of task-grouped splits +0.129 [+0.05, +0.16].
- The matched comparison chooses the operating point on held-out honest data: fair across rules, not deployable. Fit at nominal 1%, Stackelberg spends 1.08% and global 0.97%, and the nominal contrast (+0.134 [+0.073, +0.182]) is partly bought with that over-spend.

Reading. With hundreds of calibration runs per context, Stackelberg's point estimate at 1% is about +0.1 selection safety at equal realized spend, at a small uniform-TPR cost. Whether it holds beyond

the SHADE setting is not established: the evidence for a 1% advantage is a pre-specified positive result that rests on one setting with five main tasks.

7.2 Secondary results on 7 settings [E]

Δ selection safety vs global	$B = 0.5\%$	$B = 1\%$	$B = 2\%$	$B = 5\%$
global (level)	0.519	0.646	0.743	0.889
Stackelberg, equal damage ($m_0 = 10$)	+0.089 [−0.082, +0.096]	+**0.107** [+0.024, +0.152]	+0.133 [+0.074, +0.162]	+0.069 [+0.025, +0.091]
Stackelberg, unshrunk	+0.086 [−0.084, +0.090]	+0.104 [+0.021, +0.147]	+0.131 [+0.068, +0.160]	+0.067 [+0.023, +0.085]
Stackelberg, severity-weighted	+0.092 [−0.080, +0.102]	+0.124 [+0.046, +0.154]	+0.136 [+0.078, +0.168]	+0.071 [+0.026, +0.088]
partial pooling (EB n_0)	−0.219 [−0.390, −0.190]	−0.222 [−0.380, −0.157]	−0.177 [−0.287, −0.110]	−0.084 [−0.202, −0.018]
CFAR	−0.318 [−0.465, −0.241]	−0.280 [−0.460, −0.218]	−0.239 [−0.363, −0.164]	−0.143 [−0.244, −0.058]
share of cells matched (per-context rules)	0.94	1.00	1.00	1.00

Table 14: Round 4, 7 settings, in distribution, exactly spend-matched, informed context-selecting attacker; 95% intervals from the trajectory-level cluster bootstrap (the SHADE caveat above applies to every Stackelberg cell). Only the bold cell is confirmatory. Point estimates near an interval end reflect a skewed bootstrap distribution. [r4_ctx/summary.csv]

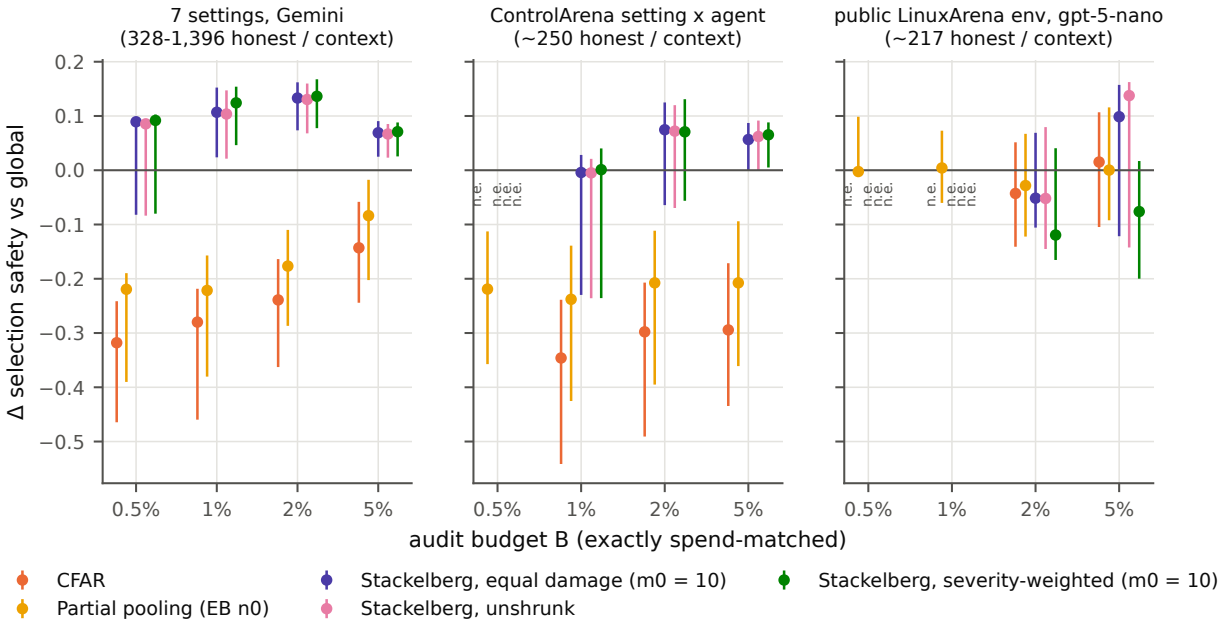


Figure 8: Round 4: Δ selection safety vs global by budget, exactly spend-matched, for 7 settings, ControlArena and public LinuxArena environments (trajectory- or task-grouped cluster bootstrap 95% intervals; missing points are not estimable). [r4_ctx/figures/]

- **Uniform attacker.** Stackelberg costs little at 1%: TPR 0.834 vs 0.854, $\Delta = -0.020$ [−0.04, +0.01]. CFAR lowers uniform TPR too (−0.047 [−0.11, −0.02]); no per-context rule raises uniform TPR at 1% on this strong-monitor panel.
- **Other attackers at 1%.** Severity-weighted Stackelberg against a damage-maximizing attacker +0.077 [+0.02, +0.10] (equal-damage +0.058 [−0.01, +0.08]); against a probing attacker with 5 free probes per context +0.067 [+0.00, +0.09] (equal-damage +0.057 [−0.01, +0.08]).
- **CFAR is the worst rule against selection at every budget** (Table 14), as in rounds 1–2.

- **Nominal fits over-spend out of sample** at $B = 1\%$ (mean realized test spend): CFAR 1.14%, Stackelberg 1.08%, pooling 1.12%, global 0.97% (plug-in thresholds).
- **Transfer.** Fit on other side tasks, Stackelberg is worse than global at every budget: 1% 0.510 vs 0.643, $\Delta = -0.133$ [−0.20, −0.08] (severity-weighted −0.108; unshrunk −0.146), 5% −0.122 [−0.18, −0.08]. Shrinkage with $m_0 = 10$ does not rescue it. Fit on other attack policies or agent models: +0.033 [−0.09, +0.10] at 1%, +0.050 [+0.01, +0.08] at 5%. The gain needs attack data from the deployed side-task distribution.

7.3 Panels with fewer honest runs per context [E]

- **ControlArena setting × agent** (≈125 calibration runs per context): the CFAR / Stackelberg spend floor is 0.63% mean, 1.30% p95; matched in 85% of cells at 1% and 41% at 0.5% (*n.e.*). At 1%: Stackelberg −0.004 [−0.23, +0.03], CFAR −0.35 [−0.54, −0.24]. At 5%: Stackelberg +0.057 [−0.00, +0.09], severity-weighted +0.065 [+0.01, +0.09]. Transfer to unseen side tasks: −0.14 at 1%, −0.23 at 5%.
- **Public LinuxArena by environment** (≈110 calibration runs per environment): per-context rules are *n.e.* at 1% (matched in 68%; floor 0.83% mean, 2.27% p95) and at 0.5% (40%). The monitor is weak at these budgets (global uniform TPR 0.17 at 1%, 0.41 at 5%); at 5% no contrast is significant (Stackelberg +0.10 [−0.12, +0.16]).
- Global and pooling have no material floor ($\leq$ 0.09% mean) and are matchable everywhere. CFAR and all Stackelberg variants share one floor (thresholds at each context's calibration maximum).

Rule of thumb. Per-context rules are usable at budget B only if $\sum_c w_c/(n_c + 1) \ll B$, i.e. at least about $1/B$ calibration runs in the smallest contexts: roughly 300 per context at 1% after a 50/50 split.[r4_ctx/tables.md]

8 What holds, what is weakened, what is dropped

Claim	Status	Evidence
Heterogeneity across deployments (τ 1.34–1.75), small within LinuxArena, none across agent models	**holds**	[t2/tables.md]
White-box probe scores are far more context-dependent than LLM scores	**holds**	honest-only; [fixes/corrections.csv (M5)]
CFAR is the worst rule against context selection (7 settings 1%, ControlArena 5%; round 4: every budget)	**holds**	independent attacker info: [audit/repro_t2.csv]; round 4 −0.280 at 1%
A few alarm-feedback probes suffice to exploit CFAR; Stackelberg is flat	**holds**	[r2_t2/tables.md (E4a)]
Stackelberg > global against selection at 5% (7 settings)	**holds**	+0.076 [+0.026, +0.124] (conformal); round 4 +0.069 [+0.025, +0.091]; [fixes2/headline_threshold_rerun.csv]
Stackelberg > global against selection at 1% (7 settings), round 4 primary [C]	**weakened**	+0.107 [+0.024, +0.152] (trajectory resampling, criterion met); main-task resampling [−0.003, +0.251]; driven by SHADE; [audit3/ctx_primary_checks.csv]

Claim	Status	Evidence
Round-4 exact spend matching as a deployment comparison	weakened	operating point chosen on test honest data (fair, not deployable); nominal Stackelberg over-spends (1.08% vs 0.97%)
Stackelberg needs on-distribution side-task attacks (round 4 transfer, 1%: -0.133)	holds	[r4_ctx/summary.csv]
Per-context rules need $\gtrsim 1/B$ calibration runs per context (ControlArena, public LinuxArena *n.e.* at 1%)	holds	[r4_ctx/tables.md]
Stackelberg > global at 1% on ControlArena	dropped	over-spend; matched 0.610 vs 0.607
Plug-in per-context quantiles over-spend with small n; Mondrian silent; pooled_cv near budget	holds	[t2/tables.md]
CFAR / max-detect raise uniform TPR for weak monitors (round 1, MB clusters)	weakened	rows partly clamped (m_ok 0.34–0.70)
CFAR breaks at $\varepsilon \approx B$; global at $\varepsilon \approx B/w_t$	holds	[t2/tables.md]
OS-CFAR (LOCO) holds 0.77–0.86 under poisoning	weakened	per panel 0.62–0.81 at 10%; worst target 0.07–0.45; [fixes/corrections.csv (m3)]
OS-CFAR (LOCO) as the recommended poisoning defense	dropped	breaks under spread poisoning (E2)
Frozen-reference tail robust to spread poisoning, over-spends under deployment-wide tail drift	holds	[r2_t2/tables.md (E2)]
Per-context thresholds do not transfer to a new agent; offsets + recalibrated level do	holds	[r2_t2/tables.md (E1b)]
Stackelberg fit on other side tasks "worse than global"	weakened	significant vs in-distribution only
Cell severity minimax 0.92 / 0.95 vs global 0.64 / 0.71	weakened	intervals touch or cross 0
E5: CFAR roughly doubles probe TPR at 1%	weakened	1% *n.e.*; 1.3–1.5× at equal spend; [fixes/corrections.csv (C1)]
E5: all 1% spend-matched rows	dropped	clamped; [fixes/corrections.csv (C1)]
E5: Stackelberg $\geq$ global on 14/17 signals at 5%	weakened	10/17 grouped, 0 CIs > 0; [fixes/corrections.csv (M5)]
E5 AUROCs (gpt-4.1 0.911, `fibs_fu_full` 0.905, ...)	weakened	originals 0.45–0.83; fabrication confound
Blind severity does not separate (AUROC 0.53–0.54)	holds	[r3_t2/tables.md]
Watcher tiers infeasible at 1% and lose to global at 5%	holds	non-clamping re-summary; [r4_ctx/r3t2_corrected/]
r3_t2 1% "matched" rows (per-org rules, tiers)	dropped	*n.e.* without clamping (matched 4–25%); [r4_ctx/r3t2_corrected/tables_noclamp.md]
r3_t2 mb_mini per-org rows at 5%	dropped	matched in 78% → *n.e.*
akshn_adj Watcher-permissive / strict / tier_w "matched" values	dropped	never matched (spend unreachable)
akshn_adj severity results (0.98–0.99)	weakened	label-leaking upper bound; [fixes/corrections.csv (m11)]

8.1 Errata relative to the October 22 version

1. **Summary item 3** ("Stackelberg beats global at 5%, not demonstrably at 1%"): round 4 met its pre-specified 1% criterion (+0.107 [+0.024, +0.152]), but only with trajectory-level resampling in the small-task settings; the gain is driven by SHADE (5 main tasks) and with main-task resampling the interval is [−0.003, +0.251]. The spend matching is an oracle operating point.
2. **5% Stackelberg gain**: stated as the interval $+0.03.. + 0.12$; it is +0.076 [+0.026, +0.124] (conformal re-run; the overview figure's 0.075 is a rounding of this).
3. **Watcher table**: values replaced by the non-clamping re-summary (e.g. mb_mini strict 0.278 → 0.281; mb_strong global 0.369 → 0.373); the 5% conclusions hold. The mb_mini per-org rules at 5% and several akshn_adj tier rows, previously shown as matched, are not estimable.
4. **Threshold convention**: stated once (Section 2.5); no verdict in this report changed.

8.2 Errata relative to the October 21 version

1. **Summary item 3 and the allocation table** ("Stackelberg 0.79 vs 0.64 at 1%"): the 1% gain is not established once spend-matched (interval $-0.11.. + 0.20$). The 5% gain stands. With attacker information independent of the defender's fit the 1% / 5% values are 0.775 / 0.957, not 0.791 / 0.966.
2. **ControlArena 1% Stackelberg row** (0.729, $-0.07..+0.27$): dropped. The nominal advantage is over-spending (1.38% vs 0.97%); spend-matched it is 0.610 vs 0.607, with only 66% of replicates bracketed.
3. **Allocation caption** ("the spend-matched rows remove this; their conclusions are the same"): false for Stackelberg at 1%, and the round-1 matcher clamps when its sweep misses B. Rows with `matched_ok` < 0.8 are now flagged.
4. **Weak-monitor bullet** ("nano trajectory at 5%: 0.079 → 0.106 / 0.162"): these rows have `matched_ok` 0.70 / 0.34 and max-detect realizes 5.56%; the gain is not established.
5. **Summary item 5, poisoning table and recipe step 4** ("OS-CFAR with a LOCO tail holds 0.77–0.86 and still tracks genuine drift"; "use OS-CFAR with a leave-one-context-out tail"): the range is one panel's; per panel 0.62–0.81 at $\varepsilon = 10\%$ with worst targets at 0.07–0.45, and a clean-data cost on LA task category (0.778 vs 0.914). Under poisoning spread across contexts the LOCO tail breaks at $\varepsilon \approx B$ (worst 0.16–0.18 at 5%). The recommended defense is now a frozen-reference tail with trusted refresh.
6. **"Untested: transfer across datasets; poisoning across several contexts"**: both were tested in round 2 (E1, E2).
7. **Recipe step 3** ("use Stackelberg fit on held-out attack data"): now conditional on attack data from the deployed side-task distribution and $\gtrsim$10–20 attacks per context or shrinkage (E1a, E4b).
8. **Spend line**: $17.11 was round 1 only; the track total is $38.93.
9. The October 21 version did not cover rounds 2–3. The round-2/3 log entries' E5 statements ("CFAR roughly doubles uniform TPR", "Stackelberg ≥ global on 14/17 signals") are corrected in Section 5; the round-3 log's 1% statements are limited to infeasibility.

9 Practitioner recipe

1. **Measure before allocating.** Fit the random-effects τ on per-context exceedances of a global 5–10% threshold, on honest data. If τ is not significant (LLM monitors within one deployment, LinuxArena, BashArena clusters), use one global threshold. Expect large τ for white-box probe signals and across deployments that share a monitor.
2. **Check budget feasibility.** Per-context thresholds cannot spend less than $\approx \sum_c w_c/(n_c+1)$. Run per-context rules only when that floor is well below B; in round 4 this needed roughly $1/B$ calibration runs per context (about 300 at 1% after a 50/50 split), and 110–125 per context were not enough. Otherwise use global, or partial pooling with cross-validated n_0. Always report realized spend next to any per-context TPR, and use a threshold rule with exact out-of-sample FPR (the conformal rule).
3. **If the attacker cannot choose its context** and the signal is heterogeneous (probes), per-context normalization raises detection at equal spend (`dec` probes $+0.18..+0.23$). Compare at equal realized spend, not at nominal level.
4. **If the attacker can choose its context, never use CFAR**: alarm feedback from 3–5 probes per context is enough to exploit it, and it was the worst rule against selection at every budget in every round. Use Stackelberg (damage-equalizing) thresholds only with $\gtrsim$10–20 attacks per context from the deployed side-task distribution, or shrunk toward the pooled attacks ($m_0 \approx 10$, $\lambda \approx 0.5$); otherwise use global. Its advantage is clear at 5% and suggestive at 1%. Fit on other side tasks it is worse than global. Use severity-weighted cell minimax only when per-cell damage is known.
5. **New agent model**: do not reuse per-context thresholds. Reuse the per-context offsets and recalibrate one global level on the new agent.
6. **Recalibration under a possible poisoner**: keep a frozen trusted reference; take per-context offsets from the new window and the tail from the reference; refresh the reference only through a trusted channel. Budget for 2–3× over-spend during genuine deployment-wide tail drift until the refresh. A LOCO tail is adequate only if the attacker can reach a single context. Outlier filters only slow a boiling frog.
7. **Expect within-context attack-type selection** to cost 0.2–0.4 more than allocation can recover.

Recipe for a Watcher-like product (round 3)

1. Rank actions by suspicion with one budget-matched threshold per deployment.
2. Do not hard-wire an always-block tier: measure its honest hit rate first (here 1.5% of honest trajectories, above a 1% budget).
3. Use severity as a tilt only if a validation set shows it discriminates; tune β and accept $\beta \to 0$.
4. Go per-org only after measuring org heterogeneity (τ), and then with Stackelberg (with $\geq$20 attacks per org, or shrunk), never CFAR.
5. Expect white-box probe signals to need per-context normalization against uniform attackers, and pair it with damage-equalizing allocation against selection.

10 Limitations and open questions

- **Monitor strength.** The Gemini panels are near-separable (AUROC ≈ 0.99); the Monitoring-Bench monitors are weak. Allocation effects are cleanest in between.
- **Few attacks per context** in the LinuxArena sweep (5–18) and in E5 (6–42 base attacks per cluster), so attack-fitted rules (Stackelberg, max-detect) are noisy there.
- **The spend matcher.** Round-1 and round-2 tables still use the clamping matcher; this report flags but does not re-estimate their `matched_ok` < 0.8 rows. Round 3 was re-summarized and round 4 uses the exact matcher. Re-running rounds 1–2 with the exact matcher is cheap (no API cost). Exact matching on the test half is an oracle operating point; deployable nominal fits over-spend.
- **Resampling unit.** In settings with few main tasks (SHADE, iac: 5; AgentDojo: 16) the pre-specified splits and bootstraps are by trajectory, which understates uncertainty for effects that concentrate in one such setting (the round-4 primary).
- **Attacker modeling.** The informed attacker uses its own calibration-split attacks; the oracle is biased low (winner's curse). Damage weights are synthetic except E3's adjudicated severity, which leaks the label.
- **E5 confound.** 94% of E5 attacks have LLM-fabricated tool outputs; probes may key on fabrication. Only 33 originals are clean, 1–15 per cluster.
- **Severity rater.** One rater model (gpt-4.1-mini) and one prompt. A rater that sees the side task would leak the label like akshn_adj.
- **Open**: poisoning against Stackelberg's attack-data refits; a trusted-refresh protocol and its cadence; per-context allocation for white-box probes on clean (non-fabricated) attacks at scale.
- **Negative results**: TM-CFAR is dominated; empirical-Bayes n_0 is worse than cross-validated n_0; agent model is not a useful context; side-task taxonomy weights are a poor damage proxy; severity tilt adds nothing with a blind rater.

Reproduction

Code: `tracks/t2_per_context/` (round 1), `tracks/r2_t2/` (round 2 and E5), `tracks/r3_t2/` (round 3), `tracks/r4_ctx/` (round 4 and the round-3 re-summary), `tracks/fixes/e5/` (C1 / M5 / m2 / m3 / m11 fixes), `tracks/audit2/`, `tracks/audit3/` (`ctx_*`), `tracks/fixes2/` (`ctx_*`, threshold re-run). Results: `results/t2/`, `results/r2_t2/`, `results/r3_t2/`, `results/r4_ctx/`, `results/fixes/{t2,r2_t2,r3_t2}/`, `results/audit/`, `results/audit2/`, `results/audit3/`, `results/fixes2/`. Raw replicates: `data/processed/{r2_t2_runs,r3_t2_runs}/`. Narrative: `EXPERIMENT_LOG.md` (entries "T2 per-context thresholds", "T2 ROUND 2", "T2 ROUND 3", "T2 ROUND 4", "Audit fixes", "Second-pass audit", "Third-pass audit", "Threshold-convention fix"). Repository: https://github.com/gregkocher/agent-monitor-detection (private).